PYTHON PROGRAMMING
ASSIGNMENT NO. = 2

1. Even or Odd Function

Write a function is_even(n) that returns "Even" if the number is even; otherwise, it returns "Odd".

Python Code

```
def is_even(n):
    if n % 2 == 0: return
        "Even"
    return "Odd"

print(is_even(8))
print(is_even(5))
```

Output Screenshot (captured sample run)



2. Largest of Two Numbers

Write a function largest(a, b) that returns the larger number.

Python Code

```
def largest(a, b): if a >= b:
        return a return b

print(largest(12, 7))
print(largest(4, 9))
```

Output Screenshot (captured sample run)



3. Print Numbers

Write a function print_numbers(n) that prints numbers from 1 to n using a loop.

Python Code

```
def print_numbers(n):
    for number in range(1, n + 1): print(number, end="
        ")
    print()

print_numbers(5)
```

Output Screenshot (captured sample run)



4. Sum of Numbers

Write a function sum_n(n) that returns the sum of numbers from 1 to n.

Python Code

```
def sum_n(n):
    total = 0
for number in range(1, n + 1): total += number
return total

print(sum_n(5))
```

Output Screenshot (captured sample run)



5. Multiplication Table

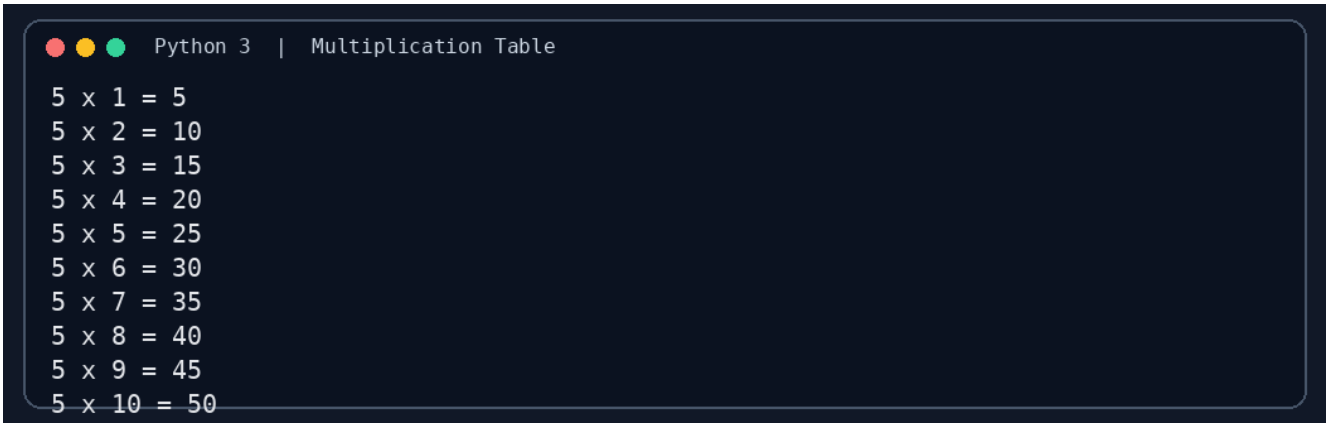
Write a function table(n) that prints the multiplication table of n from 1 to 10.

Python Code

```
def table(n):
    for i in range(1, 11): print(f"{n} x {i} = {n * i}")

table(5)
```

Output Screenshot (captured sample run)



6. Count Even Numbers

Write a function count_even(numbers) that takes a list and returns how many even numbers it contains.

Python Code

```
def count_even(numbers): count = 0
    for number in numbers: if number
        % 2 == 0:
            count += 1 return
    count

print(count_even([1, 2, 3, 4, 6]))
```

Output Screenshot (captured sample run)



7. Find Maximum

Without using max(), write a function that returns the largest number in a list.

Python Code

```
def find_max(numbers):
    if len(numbers) == 0:
        raise ValueError("The list must not be empty")

    largest_number = numbers[0] for number in
    numbers[1:]:
        if number > largest_number: largest_number
            = number
    return largest_number

print(find_max([12, 5, 27, 8, 19]))
```

Output Screenshot (captured sample run)



Note: An empty list has no largest element, so this function raises ValueError for an empty list.

11. FizzBuzz

Print numbers from 1 to 100. Print "Fizz" for multiples of 3, "Buzz" for multiples of 5, and "FizzBuzz" for multiples of both.

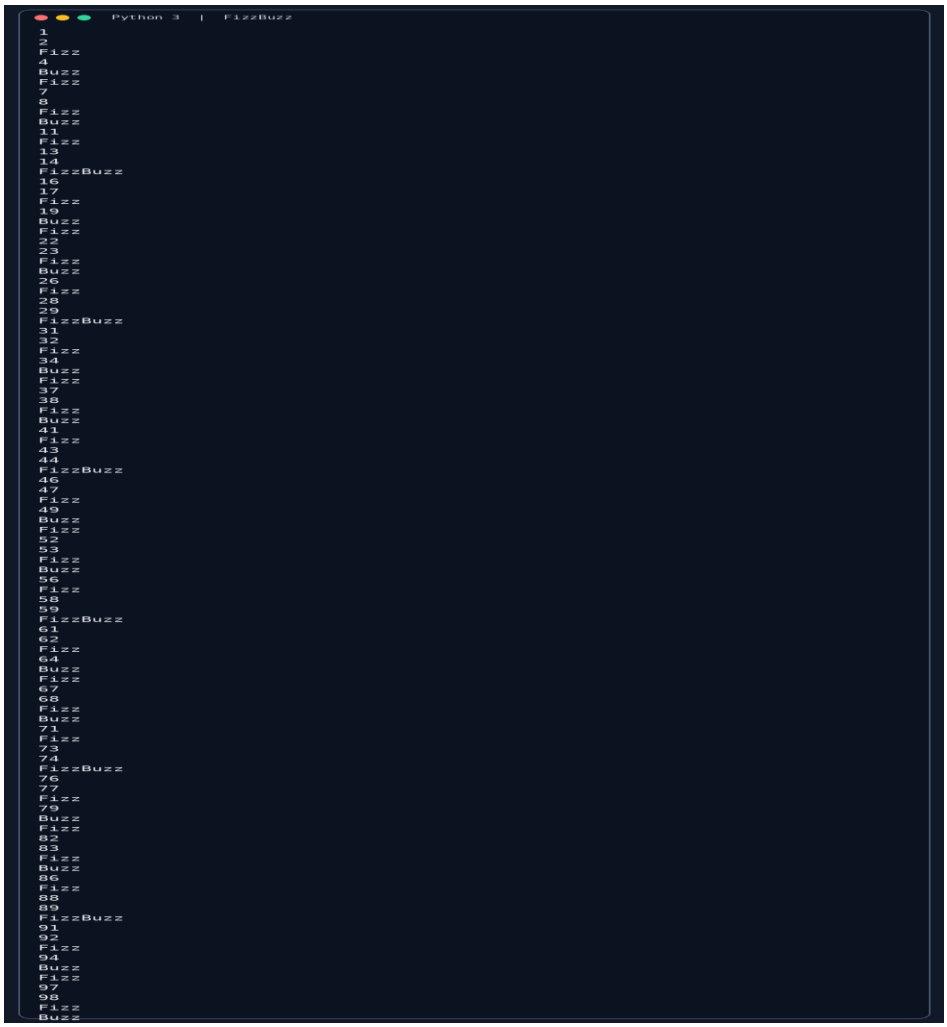
Python Code

```
def fizz_buzz():
    for number in range(1, 101):
        if number % 3 == 0 and number % 5 == 0: print("FizzBuzz")
        elif number % 3 == 0: print("Fizz")
        elif number % 5 == 0: print("Buzz")
        else:
            print(number)

fizz_buzz()

# Run the program to see all 100 lines.
```

Output Screenshot (captured sample run)



12. Palindrome Checker

Return True if a string is a palindrome. This version ignores spaces, punctuation and letter case.

Python Code

```
def is_palindrome(text): cleaned = ""
    for character in text.lower(): if
        character.isalnum():
            cleaned += character return cleaned
    == cleaned[::-1]

print(is_palindrome("Madam"))
print(is_palindrome("Hello"))
```

Output Screenshot (captured sample run)



13. Count Vowels

Write a function that counts the vowels (a, e, i, o, u) in a string.

Python Code

```
def count_vowels(text): vowels =
    "aeiou" count = 0
    for character in text.lower(): if character in
        vowels:
            count += 1 return
    count

print(count_vowels("Education"))
```

Output Screenshot (captured sample run)



14. Second Largest Number

Return the second largest distinct number in a list without using sort().

Python Code

```
def second_largest(numbers): largest = None
    second = None

    for number in numbers:
        if largest is None or number > largest: if number != largest:
                second = largest largest =
            number
        elif number != largest and (second is None or number > second): second = number

    if second is None:
        raise ValueError("At least two distinct numbers are required") return second

print(second_largest([10, 5, 8, 10, 3]))
```

Output Screenshot (captured sample run)



15. Guessing Game

Generate a random number from 1 to 10 and keep asking until the user guesses correctly.

Python Code

```
import random

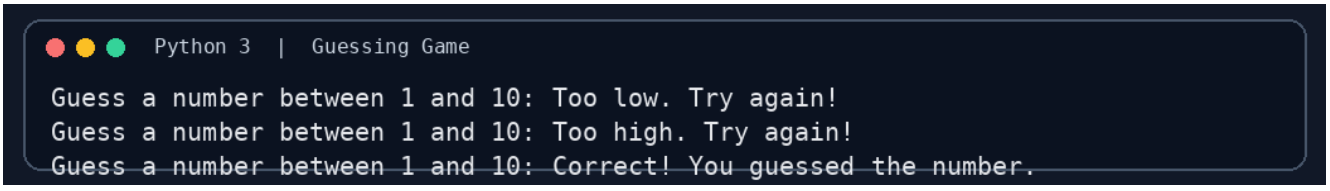
def guessing_game():
    secret_number = random.randint(1, 10) guess = 0

    while guess != secret_number:
        guess = int(input("Guess a number between 1 and 10: "))

        if guess < secret_number: print("Too low. Try
            again!")
        elif guess > secret_number: print("Too high. Try
            again!")
        else:
            print("Correct! You guessed the number.")

guessing_game()
```

Output Screenshot (captured sample run)



16. Grade Function

Return A for marks >= 90, B for marks >= 80, C for marks >= 70, D for marks >= 60, and F otherwise.

Python Code

```
def grade(marks):  if marks
        >= 90:
                return "A" elif
        marks >= 80:
                return "B" elif
        marks >= 70:
                return "C" elif
        marks >= 60:
                return "D" else:
                return "F"

print(grade(95))
print(grade(82))
print(grade(74))
print(grade(63))
print(grade(40))
```

Output Screenshot (captured sample run)

